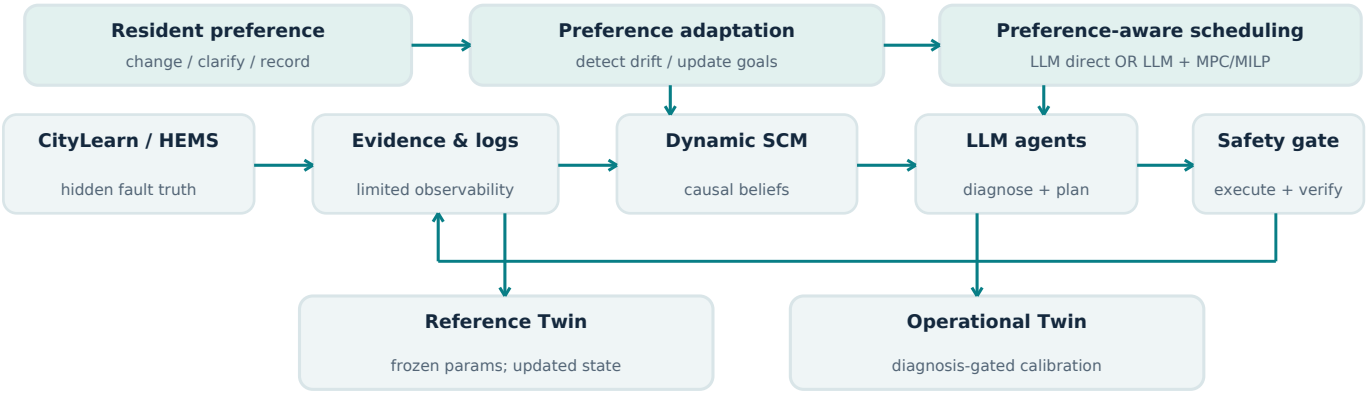
# Digital-Twin-Guided Diagnostic Adaptation for Fast and Reliable Preference-Aware Home Energy Management Using LLM-Based Multi-Agent Systems

ARTICLE | Research specification v5.0 | Final research title | 10 Nov 2026

## 01 Research objective and scientific boundary

Goal. Detect residential HEMS performance degradation, discriminate competing or simultaneous causal mechanisms under limited observability, and adapt to changing resident preferences before selecting safe, adaptive, multi-step interventions using Digital Twins and LLM agents. The claimed advance is the coupling of decision-oriented causal diagnosis to intervention choice and verified recovery, not the mere combination of DT + LLM. Novelty and superiority remain unverified hypotheses.

## 02 Reference architecture and explicit preference adaptation




Evaluator-only ground truth never enters SCM, twins, agents or preference updates.

Three agent topologies: Centralized (one agent); Hierarchical reference (Orchestrator + Diagnostic Analyst + Intervention Planner); Decentralized (Evidence / Planning / Review peers with agreement and timeout protocol). A shared preference-adaptation loop captures/clarifies user goals, detects true-vs-registered drift, updates authorized comfort/scheduling constraints and verifies preference satisfaction; no automatic DRL reward retraining. Identical SCM, twins, tools and safety across variants. Two energy-control modes: direct constrained LLM scheduling vs LLM-orchestrated MPC/MILP. DRL is not the main controller; it remains a literature baseline option.

## 03 Exact simulation benchmark (audit-reported; not revalidated here)

| Component | Selected specification |
|---|---|
| Source | CityLearn 3.2.0; audited commit fd86484b8b438f03eb40c3ec06dc90caa349842d |
| Dataset / building | ca_alameda_county_neighborhood / resstock-amy2018-2021-release-1-102040; 8,760 hourly samples |
| Thermal model | LSTM-based action-sensitive HVAC dynamics; 13-step lookback; full-year autosizing, then freeze ratings |
| Equipment | Heating 7.0294 kW; cooling 3.1479 kW; PV 10.08 kW; battery 9.9 kWh / 1.61 kW |
| Synthetic extensions | EV 40 kWh, charger 7.2 kW, arrival SOC 30%, departure target 80%, home 18:00-08:00; TOU tariff 0.35 (16:00-21:00), else 0.18 currency units/kWh |
| Reference checks | Earlier audit reported 24 checks and max 72-step temperature contrasts of 6.25 C (heating) / 8.73 C (cooling). These indicate action sensitivity, not physical accuracy. |

Artifact caveat: An audited custom_alameda package was reported in prior work, but the original files/weights and checksums are not verified available in the current workspace. Synthetic EV and tariff inputs must never be presented as measured household data.

# Causal Core, Decision Logic & Execution

Frozen architectural choices | Formal inference algorithm and numeric thresholds remain open

## 04 Hierarchical, dynamic causal model

| Family | Origin vs mechanism / diagnostic implication |
|---|---|
| Preference | A true preference change or registration gap -> mismatch between actual and registered goals; clarify/authorize the new preference, update scheduling constraints, replan and verify satisfaction. Not a physical fault. |
| Forecast | Forecast process disturbance -> error against subsequently realized load/weather; never use future observations early. |
| Model | Intrinsic wrong Twin parameters/structure versus environment-induced mismatch. A prediction residual is a shared symptom, not proof of a model fault. |
| Equipment | Real capacity/efficiency degradation -> altered effective actuation and physical response. |
| Coordination | Command delay/drop/duplication/conflict -> discrepancy between requested, approved, delivered and effective actions. |

Time-indexed causal chain: origin event(s) -> mechanisms / mediators -> sensor observations -> cost, comfort and constraint outcomes. Causes may co-occur, cascade and interact; distinguish root origin, mediator, and performance contributor. Separate latent world state from diagnostic belief. Physical evaluator/fault labels are evaluator-only; the SCM and LLM see only permitted time-stamped evidence.

## 05 Probabilistic diagnosis, identifiability and active tests

Belief: P(joint cause configuration at t | observations and intervention history up to t). Track provenance, missingness, dependence, uncertainty, and Unknown / Abstain. Only the causal engine supplies calibrated posteriors; LLMs must not rewrite them. Two causes are not identifiable if all permitted tests produce indistinguishable observation distributions under the model. Hard pairs include Model vs Equipment and Equipment vs Coordination.

| Test | Role / safety boundary |
|---|---|
| L0 Passive | Log, preference, forecast, action and residual review; read-only. |
| L1 Virtual | Isolated Twin hypotheses; model-conditional, not independent causal proof. |
| L2 Extra evidence | Request existing telemetry/acknowledgment or user clarification; availability not assumed. |
| L3 Physical probe | Limited time-bound actuation; authorized by independent safety checks, with comfort/energy effects modeled. |

Probe decision: use a test only if its expected decision improvement justifies cost, delay, physical risk and model uncertainty. Information gain alone is insufficient. Physical probes alter future state and require sequential evaluation. Counterfactuals require an explicit SCM, inferred latent conditions, intervention semantics and uncertainty; a Twin parameter sweep alone is sensitivity analysis.

## 06 Intervention contract and safety policy

Typed action library: D1 inspect_evidence; D2 request_telemetry; D3 run_twin_test; D4 request_safe_probe; C1 clarify_preference; C2 update_forecast; C3 calibrate_twin; C4 replan_energy; C5 derate_equipment; C6 recover_coordination; C7 activate_fallback. Plans are conditional, multi-step, versioned, bounded and rollback-aware; no arbitrary code or direct unsafe actuation.

Four gates: (1) schema/type/dependency validation; (2) hard safety, authorization, idempotency and command-delivery checks; (3) isolated Twin/causal scenario evaluation under uncertainty; (4) actual runtime recovery verification. Hybrid execution: batch only compatible, non-mutating operations; execute state-changing steps sequentially, reassess after each. Missing acknowledgment != non-delivery; no blind retry. Reference Twin parameters remain fixed; Operational Twin calibration requires evidence, isolated validation and rollback.

# Literature & Evaluation

Core references, reproducible baselines and evidence requirements

## 07 Core literature and appropriate experimental roles

| Paper / DOI | Role in our study |
|---|---|
| Du et al. (2026), LA-UPAHEM \| 10.1016/j.apenergy.2026.127976 | Core + adapted control baseline: preference-to-state/reward, iterative LLM-DRL; reported 2.5-3.5h adaptation motivated alternative. |
| Li et al. (2026), DARC \| 10.3390/en19163817 | Core + conditional repair baseline: constraint diagnosis, checker/critic and iterative plan repair; campus microgrid task differs. |
| Huang (2024) \| 10.1016/j.enbuild.2024.114378; Gao (2024) | Core diagnosis + Bayesian baseline: causal/Bayesian HVAC FDD; model semantics and task must be adapted transparently. |
| Li et al. (2026), Agentic AFDD \| 10.1016/j.buildenv.2025.114104 | Core + rule-based FDD baseline: agent, ontology and diagnostic rules. |
| El Makroum et al., Agentic AI HEMS \| 10.1016/j.rineng.2026.109857 | Core + adapted scheduling baseline: LLM multi-agent scheduling and orchestrator/device roles. |
| Jia et al. (2026), STAR \| 10.1016/j.enbuild.2026.117763 | Core / collision / helper: DT anomaly isolation, synchronization and optimization; diagnostic baseline only if tasks match. |
| Ren; Aghdam; Wu; Zhang; Hwang; Barricelli; Wang; Rezaei | Helpers / collision checks: safe DT-MARL, residential DT, preference authorization/elicitation, DT information fusion, smart-home LLM/DT, agentic maintenance, energy-agent surveys. Verify exact citations from source spreadsheet. |

## 08 Experimental evidence required

HEMS control baselines: rule-based; MPC/MILP; constrained LLM schedule; LA-UPAHEM/El Makroum adaptations on matched tasks. Diagnosis baselines: residual threshold; rule-based FDD; Bayesian/causal FDD. Intervention ablations: no SCM, no active probing, no Twin counterfactual, single-step vs conditional multi-step, fixed-rule and DARC-inspired repair. Full comparisons: 3 agent topologies x 2 control modes with matched access, resources and safety.

Evaluate independently: (i) diagnosis: multilabel F1, joint-cause, calibration, false root attribution, abstention, identifiability; (ii) causal predictions: intervention/counterfactual error on paired simulator episodes, assumption sensitivity; (iii) probing: decision value, unnecessary tests, risk/cost; (iv) recovery: time, regret vs evaluator-only oracle, comfort/cost, hard violations, recalibration mistakes, latency/tokens. Preserve matched seeds, holdouts and confidence intervals.

## 09 Project phase map

| Phase | Deliverable |
|---|---|
| P0 / P1 | Research contracts; verify CityLearn commit, recover audited custom_alameda, load building/LSTM; validate sizing, EV, tariff, thermal/energy behavior. |
| P2 / P3 | Five-family hidden-label fault benchmark including combined causes; comparable HEMS controllers and adapted literature baselines. |
| P4 / P5 | Reference/Operational Twins; dynamic causal graph, belief inference, identifiability and selective safe probing. |
| P6 / P7 | Conditional intervention planning, four safety gates and hybrid execution; three LLM agent topologies. |
| P8 / P9 | Matched experiments, ablations, robustness, statistics; novelty audit, manuscript and reproducibility package. |